

Definition

If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ then the dot product of $\vec{a}$ and $\vec{b}$ is given by

$$\vec{a} \cdot \vec{b} = \langle a_1, a_2, a_3 \rangle \cdot \langle b_1, b_2, b_3 \rangle$$
$$= \underbrace{a_1 b_1 + a_2 b_2 + a_3 b_3}_{\text{Number}}$$

Examples:

$$(1) \quad \langle -2, 3 \rangle \cdot \langle 4, -5 \rangle = -8 - 15 = -23$$

$$(2) \quad (8\vec{i} - 2\vec{j} + 3\vec{k}) \cdot (2\vec{i} + 4\vec{k}) = 16 + 0 + 12 = 28$$

Properties of the dot product p801

$\vec{a}$, $\vec{b}$ and $\vec{c}$ vectors in $\mathbb{R}^n$ and $c \in \mathbb{R}$

$$(1) \quad \vec{a} \cdot \vec{a} = |\vec{a}|^2$$

$$(2) \quad \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

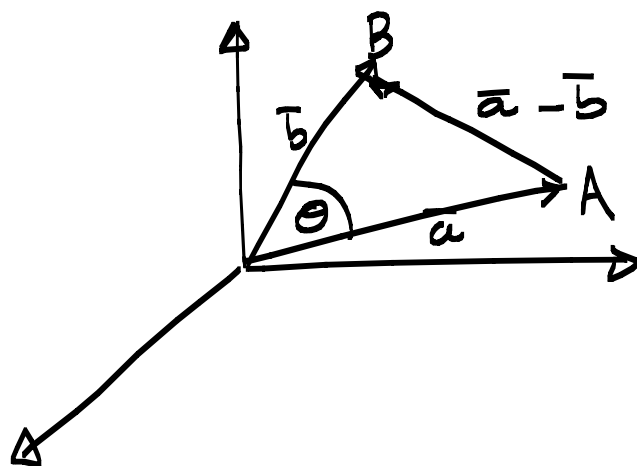
$$(3) \quad \vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

$$(4) (c\bar{a}) \cdot \bar{b} = c(\bar{a} \cdot \bar{b}) = \bar{a} \cdot c\bar{b}$$

$$(5) \bar{0} \cdot \bar{a} = \bar{0}$$

Definition

The angle θ between $\bar{a}$ and $\bar{b}$
where $0 \leq \theta \leq \pi$



If θ is the angle between $\bar{a}$ and $\bar{b}$
then $\bar{a} \cdot \bar{b} = |\bar{a}| |\bar{b}| \cos \theta$ p 801

$$\text{OR} \\ \cos \theta = \frac{\bar{a} \cdot \bar{b}}{|\bar{a}| |\bar{b}|}$$

Examples:

① Determine the angle between $\bar{a} = \langle -1, 1 \rangle$
and $\bar{b} = \langle 1, 1 \rangle$

$$\cos \theta = \frac{-1 + 1}{\sqrt{2} \cdot \sqrt{2}} = 0$$

$$\Rightarrow \theta = \arccos 0 = \frac{\pi}{2} \quad (0 \leq \theta \leq \pi)$$

(2) Determine the angle between

$$\vec{a} = \langle 1, -2, 3 \rangle \text{ and } \vec{b} = \langle -3, 2, 1 \rangle$$

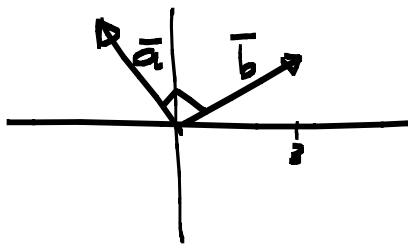
$$\cos \theta = \frac{-3 - 4 + 3}{\sqrt{14} \sqrt{14}} = -\frac{4}{14}$$

$$\Rightarrow \theta = \arccos\left(-\frac{2}{7}\right)$$

Definition Two non zero vectors $\vec{a}$ and $\vec{b}$ is orthogonal (perpendicular) if the angle between them is $\theta = \frac{\pi}{2}$ that is $\vec{a} \cdot \vec{b} = 0$

Example: ① $\vec{a} = \langle -1, 3 \rangle$ and $\vec{b} = \langle 3, 1 \rangle$

Check that $\vec{a} \cdot \vec{b} = 0$



(2) Determine k so that $\vec{a} = \langle -1, 3, k \rangle$ and $\vec{b} = \langle 2, k, 3 \rangle$ are orthogonal.

$$\vec{a} \perp \vec{b} \Rightarrow \langle 2, k, 3 \rangle \cdot \langle -1, 3, k \rangle = 0$$

$$\Rightarrow -2 + 6k = 0$$

$$\Rightarrow \underline{k = \frac{1}{3}}$$